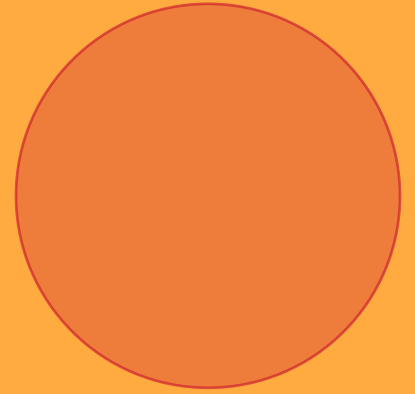


Does Advanced Metrics Predict NBA Win %?



Abstract

This study examines whether Net Rating, True Shooting Percentage (TS%), and Pace can predict NBA team win percentage using 2023–24 regular season data for 26 teams (Basketball Reference).

A multiple linear regression found that Net Rating ($\beta = 0.022$, $p < 0.001$) and TS% ($\beta = 1.419$, $p = 0.027$) are statistically significant predictors of winning, while Pace was not ($p = 0.234$).

The model explains 96.7% of the variance in win percentage ($R^2 = 0.967$), confirming that offensive and defensive efficiency drives NBA team success rather than tempo/pace.

Data Context

Source

Basketball Reference
basketball-reference.com
Advanced Stats table
2023–24 NBA Season

Collection

Derived from official NBA
box scores via standardized
formulas. No experiment —
observational study.

Cases

26 NBA team-season
observations from the
2023–24 regular season
(82 games each)

Variables

RESPONSE VARIABLE (Dependent)

Win Percentage

Wins $\div$ 82 Games | Numerical (continuous) | Range: 0.305 – 0.780 in 2023–24

EXPLANATORY VARIABLES (Independent)

Net Rating

Offensive Rating minus Defensive Rating per 100 possessions. Captures efficiency margin.

Numerical (continuous)

True Shooting %

Shooting efficiency incorporating 2-pointers, 3-pointers, and free throws.

Numerical (continuous)

Pace

Estimated possessions per 48 minutes. Measures team tempo.

Numerical (continuous)

Research Question

Do Net Rating, True Shooting Percentage, and Pace significantly predict NBA team regular season win percentage and which metric is the strongest predictor?

Table 1: Summary Statistics

Variable	Mean	Std Dev	Min	Median	Max
Win Percentage	54.6%	12.2%	30.5%	57.3%	78.0%
Net Rating	1.49	4.46	-7.1	2.35	11.6
True Shooting %	58.4%	1.7%	54.3%	58.4%	60.9%
Pace	98.4	1.69	95.2	98.3	101.7

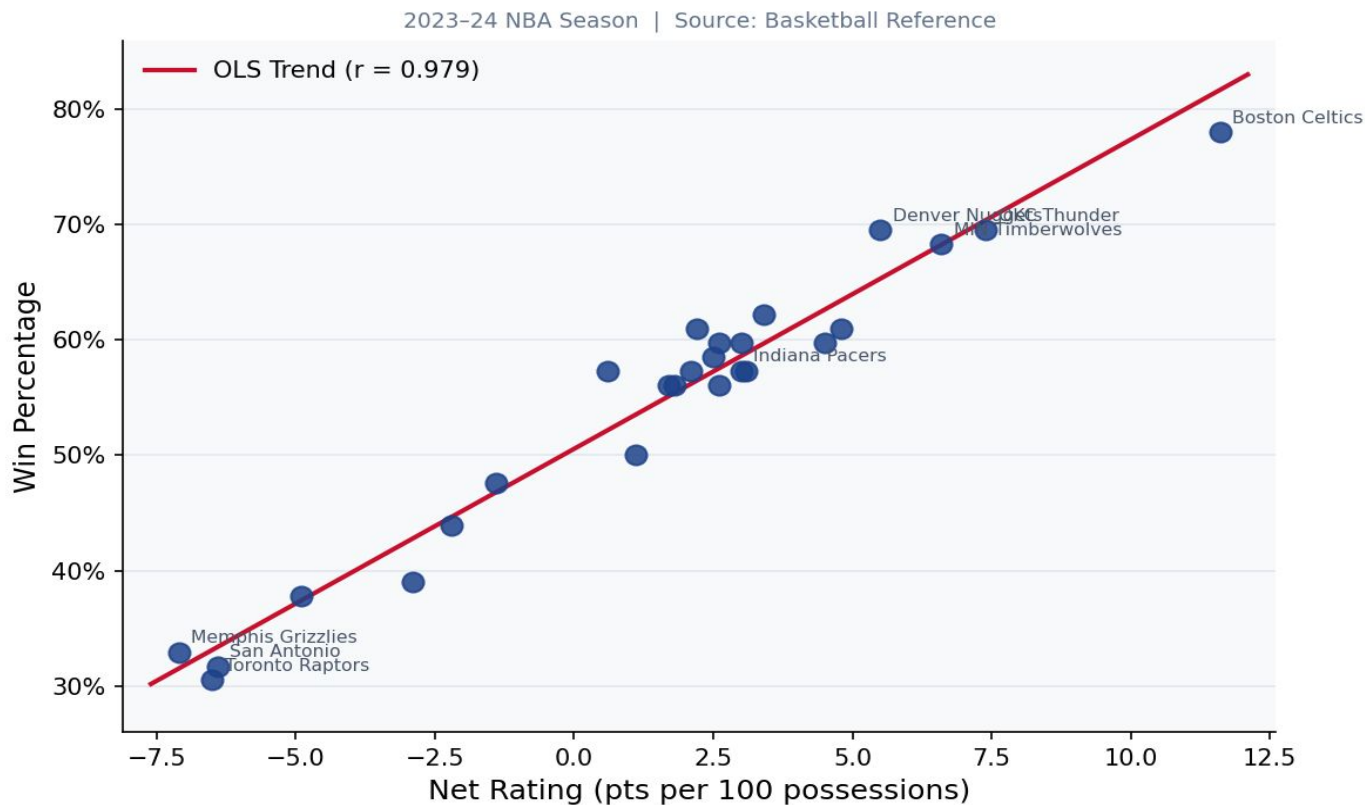
n = 26 team-season observations | Source: Basketball Reference, 2023–24 NBA Advanced Stats

r = 0.979
NRtg × Win %

r = 0.792
TS% × Win %

**r =
-0.276**
Pace × Win %

Figure 1: Net Rating vs. Win Percentage



$r^2 = 0.979$

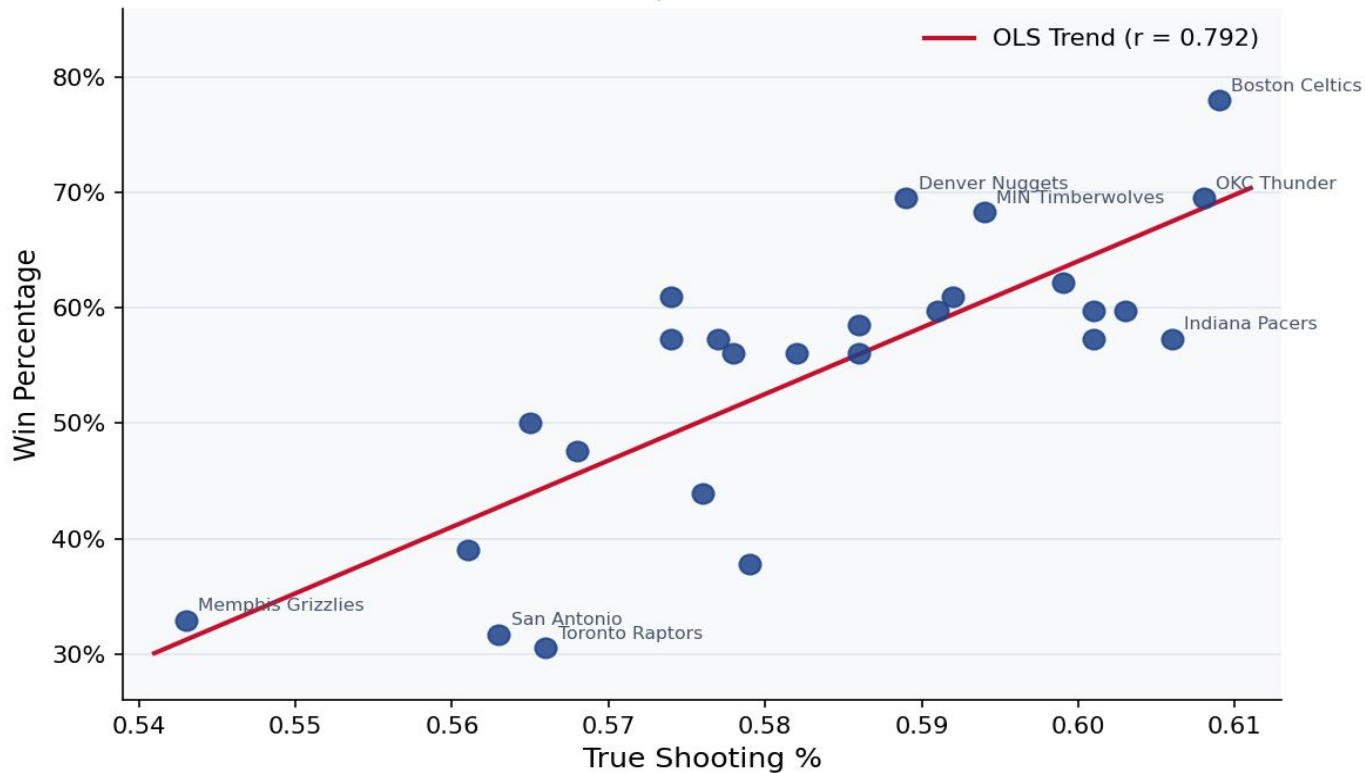
Strongest
predictor

**Each +1 NRtg
 $\approx$ +2.2% Win %**

$p < 0.001$

Figure 2: True Shooting % vs. Win Percentage

2023-24 NBA Season | Source: Basketball Reference



$r^2 = 0.792$

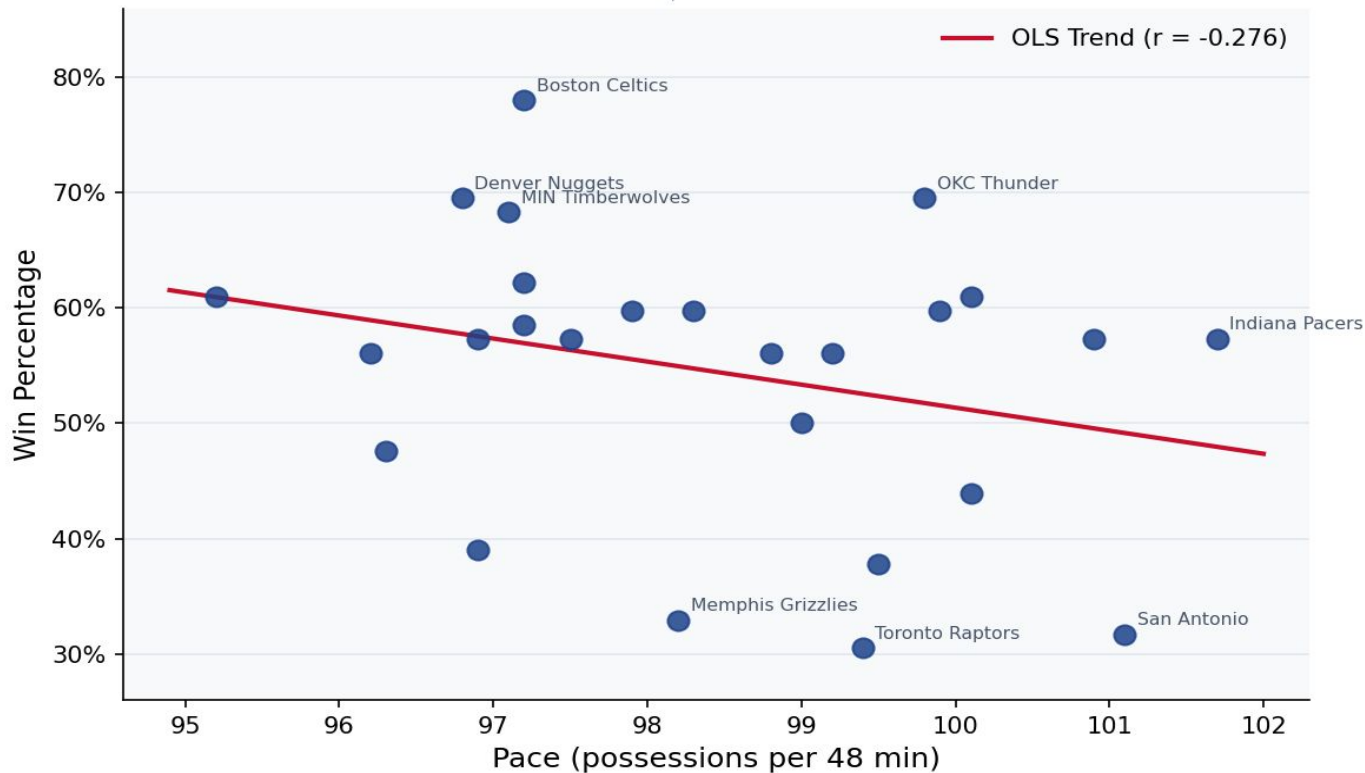
Moderate
correlation

$\beta = 1.419$
Significant

$p = 0.027$

Figure 3: Pace vs. Win Percentage

2023-24 NBA Season | Source: Basketball Reference



$$r^2 = -0.276$$

Weak
correlation

$$\beta = -0.005$$

Not Sig.

$$p = 0.234$$

Figure 4: Correlation Matrix

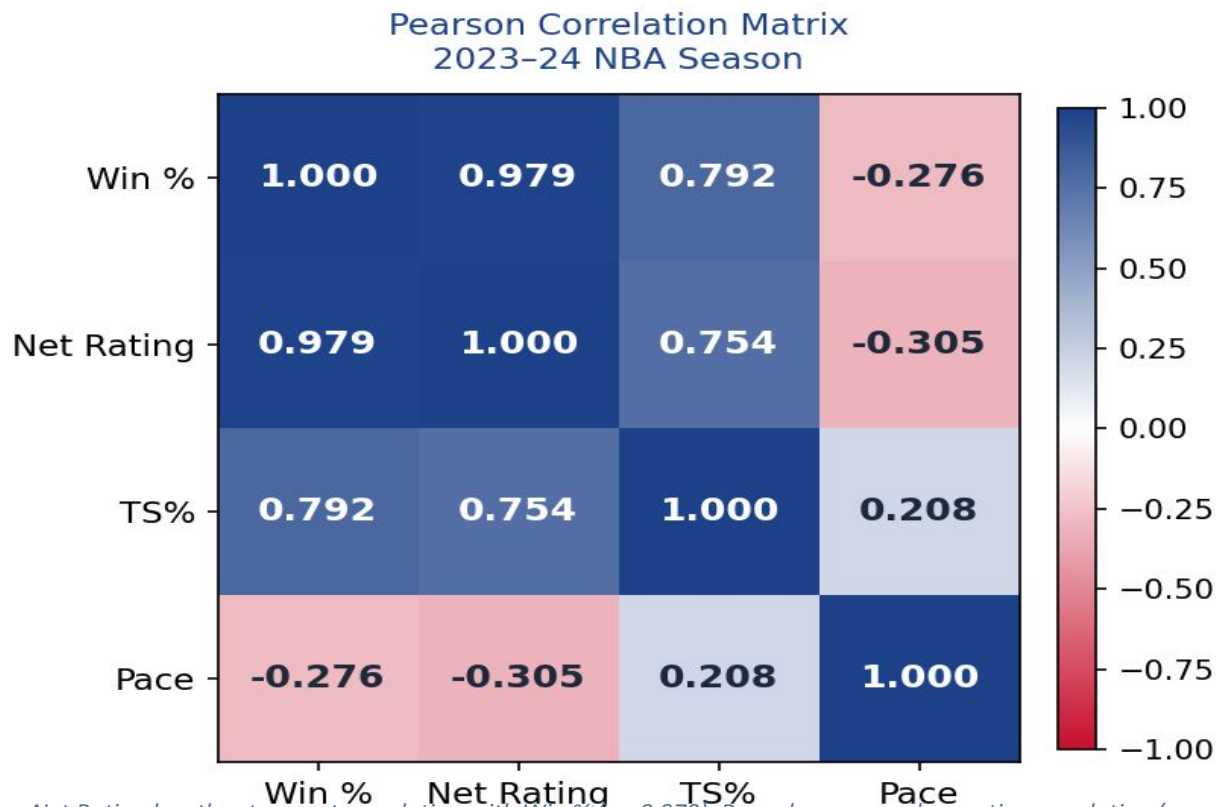


Table 2: Regression Results

win_pct ~ NRTg + TS_pct + Pace

Predictor	Estimate (β)	Std. Error	t-value	p-value	Sig.
Intercept	0.1777	0.295	0.60	0.554	ns
Net Rating	0.0222	0.0023	9.607	< 0.001	***
True Shooting %	1.4187	0.598	2.373	0.027	*
Pace	-0.0050	0.0041	-1.222	0.234	ns

*** $p < 0.001$ * $p < 0.05$ ns = not significant

H₀: $\beta = 0$ (predictor has no linear relationship with win %) **H_a:** $\beta \neq 0$ (predictor has a significant linear relationship)

Model Fit & R²

$$R^2 = 0.967$$

96.7% of variance
in win % explained
by the model

Exceptional fit

Net Rating ($\beta = 0.022$, $p < 0.001$)

Dominant predictor. Each +1 NRtg $\rightarrow$ ~2.2% more wins. Boston's +11.6 NRtg = 78% win rate.

True Shooting % ($\beta = 1.419$, $p = 0.027$)

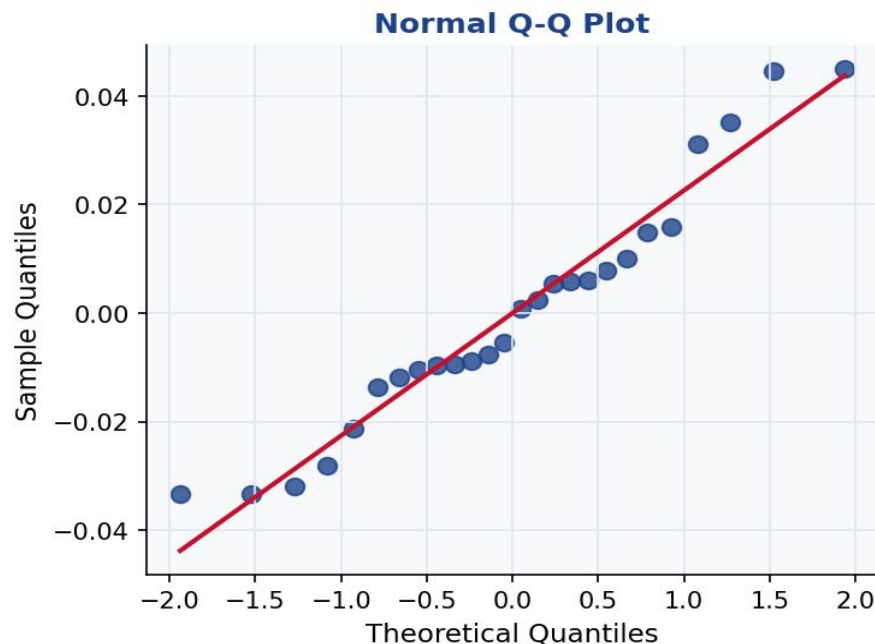
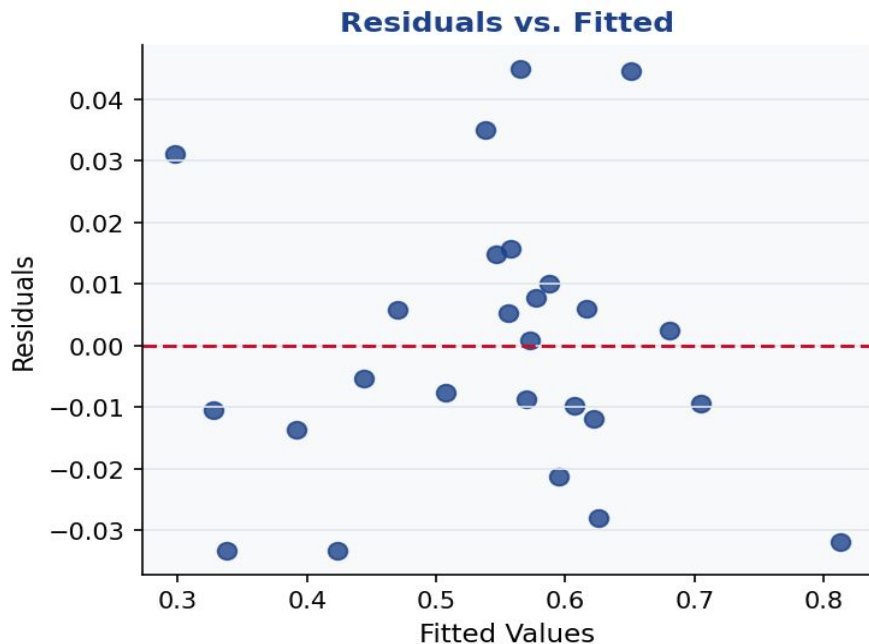
Significant secondary driver. Shooting efficiency predicts wins beyond Net Rating alone.

Pace ($\beta = -0.005$, $p = 0.234$)

Not significant. Playing faster does seem to predict winning.

Figure 5: Regression Diagnostics

Regression Diagnostics | $\text{win_pct} \sim \text{NRtg} + \text{TS_pct} + \text{Pace}$



Residuals vs. Fitted: points scatter randomly around zero — no clear pattern, confirming linearity and homoscedasticity.

Normal Q-Q Plot: points fall close to the diagonal line, supporting the normality of residuals assumption.

Conclusion

Answer to the Research Question

Net Rating ($p < 0.001$) and TS% ($p = 0.027$) significantly predict win %. Pace does not ($p = 0.234$). Efficiency drives winning — not tempo.

Why It Matters

96.7% of win % variance explained by three metrics. Front offices prioritizing efficiency are statistically validated by this analysis.

Practical Implication

Boston's NRtg of +11.6 translates directly to their 78% win rate. Efficiency is the clearest statistical path to winning in the modern NBA.